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Measuring Attitudes toward AI's Impact on the Job Market in Bangladesh: A Study at Pabna University of Science and Technology

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Abstract. Artificial Intelligence (AI) is changing work and raising practical questions about job displacement. We set out to measure how a university community in Bangladesh views these changes and to identify who feels most at risk or most ready to adapt. We ran a field survey at Pabna University of Science and Technology using stratified random sampling. 370 of 4,795 completed a 33-item questionnaire on AI familiarity, displacement belief, job concern, and adaptation confidence. Reliability was good (Cronbach's $\alpha = 0.813$). The sample was 70.8% aged 15–24, 74.1% male, and 55.7% urban. Univariate results showed that 52.2% think AI could displace jobs and 64.3% are likely or very likely to take training. Bivariate tests and correspondence analysis showed: undergraduates report lower AI familiarity than faculty and staff; younger groups report more worry about job security; income and discipline shape perceptions of opportunity. Ordinal logistic regressions found that age increases the odds of higher job concern ($\chi^2 = 10.314$, $p = .006$); greater AI familiarity predicts adaptation confidence ($\chi^2 = 56.645$, $p < .001$); higher adaptation confidence predicts willingness to train ($\chi^2 = 62.197$, $p < .001$); gender is not significant. Factor analysis supported the structure (KMO = 0.83; Bartlett's $\chi^2(528) = 3371.182$, $p < .001$). Nine components explained 57.486% of the variance: learning orientation; industry awareness; experience/status; confidence in AI; views on government initiatives; job-impact concerns; opportunity perceptions; discipline; residential status. Anxiety about AI is concentrated in younger and less experienced groups, while readiness to learn is strong. Priorities are discipline-specific AI literacy and reskilling, extra support for early-career groups, and closer work between universities, industry, and government. The validated instrument and nine-factor frame offer a clear baseline to track change and to test whether these actions reduce displacement risk.

Keywords: Artificial Intelligence (AI), Job displacement, Correspondence analysis, Logistic regression, Factor analysis.

Gallic acid Mediated Aqueous Production of Functionalized Bismuth Nanoparticles for Reducing Multiple bonds in Organic compounds

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Abstract: Gallic acid functionalized metal nanoparticles are promising in biomedical and catalytic applications due to their unusually large clean surface area, colloidal stability, and antioxidant properties. In this research, gallic acid functionalized bismuth nanoparticles (GA-BiNPs) were synthesized via a simple aqueous reduction of Bi^{3+} ions. Gallic acid served as both reducing and capping agent. Fourier transform infrared (FTIR) spectroscopy confirmed the formation of metallic-Bi and coordination of gallic acid onto the surface of BiNPs through ligations of O–H, and COO^- groups. DLS analysis indicated a hydrodynamic diameter of 290 nm with a polydispersity index of 0.33. SEM data revealed spherical nanoparticles with an average size of 16.87 nm, while TEM and high-resolution TEM confirmed the rhombohedral crystalline structure of GA-BiNPs visualizing lattice fringes and interplanar spacings. Five distinct rings showed in SAED pattern for individual interplanar distance, which perfectly matched with Rhombohedral polycrystalline structure. TGA-DTA data showed multistage decomposition corresponding to adsorbed water removal and ligand degradation of 40% total weight loss confirming surface functionalization and thermal stability of the Bi-core. GA-BiNPs significantly enhanced the reduction of *p*-nitro benzaldehyde at room temperature. Complete conversion of both –CHO and –NO₂ groups was achieved at 60 °C. Similarly, while NaBH₄ reduced cinnamaldehyde only to cinnamyl alcohol, the presence of GA-BiNPs enabled its complete reduction to 3-phenylpropan-1-ol under identical conditions. Overall, this study demonstrates a green and cost-effective synthesis of GA-BiNPs with superior catalytic performances for organic compound reduction.

Keywords: Chelate complexes, functionalization, auto-reduction, colloidal stability, catalysis.